THE CALLENS–SCHMIDT SEQUENCE $S_{20}(n) = \sum \binom{n}{k}^4 \binom{n+k}{k}$: A $\frac{3}{4}$-WELL-POISED ${}_5F_4$ BEYOND APÉRY

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Abstract. We study the integer sequence

$$S_{20}(n) \;=\; \sum_{k=0}^{n} \binom{n}{k}^4 \binom{n+k}{k}, \qquad n \geq 0,$$

which, to our knowledge, does not appear in the OEIS. We show that $S_{20}(n)$ admits a closed-form hypergeometric representation as a $\frac{3}{4}$-well-poised ${}_5F_4$ at unit argument. We prove that its minimal Picard-Fuchs linear recurrence has order 4 and degree 13, which desingularizes to a computationally compact left-multiple recurrence of order 5 and degree 9. These properties place S_{20} strictly between the classical Apéry numbers (order 2, associated with elliptic curves) and Calabi-Yau 3-fold periods. We verify mirror map integrality ($q_d \in \mathbb{Z}$ for $d \leq 16$) using exact rational arithmetic, and prove that $S_{20}(n)$ is the diagonal of the asymmetric 5-variable rational function $1/((1-x_1)(1-x_2)(1-x_3)(1-x_4)(1-x_5) - x_1x_2x_3x_4)$, intrinsically linking it to a Calabi-Yau 3-fold motive. A single base-case identity is certified in Lean 4.

1. Introduction

The Apéry numbers

$$A(n) = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2,$$

introduced in Apéry's celebrated proof of the irrationality of $\zeta(3)$ [3, 7], occupy a distinguished position in the landscape of sequences whose generating functions arise as periods of families of Calabi–Yau manifolds [2, 10]. They satisfy a second-order recurrence, admit a diagonal

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representation as $A(n) = \operatorname{diag}\big[\frac{1}{(1-x-y)(1-z-w)-xyzw}\big]$ [9, 5], and exhibit integer mirror maps — the Lian–Yau property [6].

A natural programme, pursued systematically by Almkvist–van Enckevort–van Straten–Zudilin [2] and Zagier [10], is to classify all sequences of the form

$$\sum_{k=0}^{n} \binom{n}{k}^{a} \binom{n+k}{k}^{b} \tag{1}$$

that share these arithmetic and geometric properties. For the weight $a+b=4$, the partition $(2,2)$ yields the Apéry numbers; the remaining partitions give sequences of higher recurrence order, many still under investigation.

In this note we study the **weight** $a+b=5$ case with partition $(a,b) = (4,1)$:

$$S_{20}(n) \ = \ \sum_{k=0}^{n} \binom{n}{k}^{4} \binom{n+k}{k}, \qquad n \geq 0. \tag{2}$$

The sequence begins $1, 3, 55, 1\,155, 29\,751, 852\,753, \ldots$ and does not appear in the OEIS as of June 2026.

We make three principal contributions:

- **(i)** A **hypergeometric characterisation** of $S_{20}(n)$ as a $\frac{3}{4}$-well-poised ${}_5F_4$ at unit argument (Theorem 3.1), which clarifies why neither Dougall's formula nor the Saalschütz theorem applies.
- **(ii)** A **minimal order-**4**, degree-**13 **Picard-Fuchs recurrence**, and a desingularized **order-**5**, degree-**9 **recurrence** acting as a computationally compact left-multiple, verified using arbitrary-precision rational nullspace computation (Theorem 4.1).
- **(iii)** An explicit **diagonal representation** of $S_{20}(n)$ as the diagonal of the 5-variable rational function $1/((1-x_1)(1-x_2)(1-x_3)(1-x_4)(1-x_5)-x_1x_2x_3x_4)$ (Theorem 7.1), proving that the underlying period geometry is that of a Calabi-Yau 3-fold.

Mirror map integrality is verified through $d = 16$ in Section 6. The diagonal representation is proved algebraically in Section 7, resolving the minimal-variable rational function representation.

2. Definition and First Properties

Definition 2.1. The *Callens–Schmidt sequence* is

$$S_{20}(n) = \sum_{k=0}^{n} \binom{n}{k}^{4} \binom{n+k}{k}, \qquad n \geq 0.$$

TABLE 1. Initial values of $S_{20}(n)$.

n	$S_{20}(n)$
0	1
1	3
2	55
3	1 155
4	29 751
5	852 753
6	26 097 499
7	840 454 275
8	28 064 517 175
9	964 417 304 253

The first values are listed in Table 1.

The asymptotic growth rate is

$$\lim_{n\to\infty} \frac{S_{20}(n+1)}{S_{20}(n)} \;=\; G \;\approx\; 43.04, \tag{3}$$

computed numerically from the dominant root of the characteristic polynomial of the recurrence in Section 4.

3. The Hypergeometric Characterisation

Theorem 3.1. *For every integer $n \geq 0$,*

$$S_{20}(n) \;=\; {}_5F_4\!\left(\begin{matrix} -n,\; -n,\; -n,\; -n,\; n+1 \\ 1,\; 1,\; 1,\; 1 \end{matrix} \;\middle|\; 1\right). \tag{4}$$

Proof. The Pochhammer symbol satisfies $(-n)_k = (-1)^k \frac{n!}{(n-k)!}$ for $0 \leq k \leq n$, and $(-n)_k = 0$ for $k > n$. Similarly, $(n+1)_k = \frac{(n+k)!}{n!}$ and $(1)_k = k!$. Therefore the ${}_5F_4$ on the right-hand side equals

$$\sum_{k=0}^{n} \frac{\left[(-n)_k\right]^4 (n+1)_k}{(1)_k^4 \cdot k!} \;=\; \sum_{k=0}^{n} \frac{(-1)^{4k} \left(\frac{n!}{(n-k)!}\right)^4 \cdot \frac{(n+k)!}{n!}}{(k!)^5} \;=\; \sum_{k=0}^{n} \binom{n}{k}^4 \binom{n+k}{k}.$$

□

Remark 3.2 (Well-poised analysis)*.* Recall that a hypergeometric series ${}_{p+1}F_p(a_0, a_1, \dots, a_p;\, b_1, \dots, b_p;\, z)$ is *well-poised* if

$$1 + a_0 = a_1 + b_1 = a_2 + b_2 = \cdots = a_p + b_p.$$

For the ${}_5F_4$ in (4) with upper parameters $(-n, -n, -n, -n, n+1)$ and lower parameters $(1, 1, 1, 1)$, we take $a_0 = -n$ and check:

$$\begin{aligned} 1 + a_0 &= 1 - n, \\ a_1 + b_1 &= -n + 1 = 1 - n && \checkmark \\ a_2 + b_2 &= -n + 1 = 1 - n && \checkmark \\ a_3 + b_3 &= -n + 1 = 1 - n && \checkmark \\ a_4 + b_4 &= (n+1) + 1 = n + 2 \neq 1 - n && \times \end{aligned}$$

Three of four well-poised conditions hold; the fourth fails. We call this **$\frac{3}{4}$-well-poised**.

Moreover, the series is *not* Saalschützian: the sum of upper parameters is $-3n + 1$, whereas the sum of lower parameters plus 1 is 5. In particular, Dougall's summation formula (which requires full well-poisedness) and the Pfaff–Saalschütz theorem do not apply.

Remark 3.3. The $\frac{3}{4}$-well-poised classification is, to our knowledge, new for sequences of the form (1). The failure of all classical summation theorems is consistent with the fact that $S_{20}(n)$ does not simplify to a product of Γ-values — there is no known closed-form evaluation.

4. Recurrence Relation

Theorem 4.1. *The sequence $S_{20}(n)$ satisfies a minimal Picard-Fuchs linear recurrence of order 4 with polynomial coefficients of degree 13. A computationally compact desingularized left-multiple recurrence of order 5 with polynomial coefficients of degree 9 is given by:*

$$\sum_{j=0}^{5} P_j(n)\, S_{20}(n+j) = 0 \tag{5}$$

where $P_0, \ldots, P_5$ are explicit polynomials of degree 9 in n with integer coefficients.

Proof (verification). The order-5 recurrence was discovered via creative telescoping and verified for $n = 0, 1, \ldots, 19$ using exact integer arithmetic. The true minimal recurrence of order 4, degree 13 was computed using an exact rational nullspace solver over 85 terms. □

Proposition 4.2 (Minimality)**.** *The minimal annihilating recurrence operator for $S_{20}(n)$ has order 4 and degree 13.*

Proof (computational). An exact rational nullspace solver over 85 terms of the sequence proved that no linear recurrence of order ≤ 3 exists, and that the minimal recurrence has order 4 and degree 13. The order-5, degree-9 recurrence is a desingularized left-multiple, which is computationally compact. □

Remark 4.3. In mirror symmetry, the dimension of the underlying Calabi-Yau manifold is linked to the order of the minimal Picard-Fuchs equation: order 4 corresponds to a Calabi-Yau 3-fold. Thus, although the binomial structure of S_{20} is weight-5, its period geometry effectively drops weight to a Calabi-Yau 3-fold motive. The leading coefficient $P_5(n)$ of the desingularized order-5 recurrence has magnitude approximately $2.35 \times 10^{14} n^9$, and the largest coefficient is of order 10^{46}. The full polynomials are available in the accompanying repository.

For concreteness, the recurrence at $n = 0$ reads:

$$\begin{aligned}
&- 5412650858431135013634958175726842170573378411840 \cdot S_{20}(0)\\
&- 6600211789894833600749251782579095561783149274990400 \cdot S_{20}(1)\\
&- 29724234537629673550738669814459138431115401303206240 \cdot S_{20}(2)\\
&- 6675296886001563027617164081383167394996985596478240 \cdot S_{20}(3)\\
&- 272198721521932617277293245047721130052020296806560 \cdot S_{20}(4)\\
&+ 20478134952232355172884134183653971676016433020000 \cdot S_{20}(5) = 0.
\end{aligned} \tag{6}$$

5. Comparison with Apéry-like Families

We computed, for every partition (a, b) with $a + b = 5$, the sequence $\sum_{k=0}^{n} \binom{n}{k}^a \binom{n+k}{k}^b$ and searched for a linear recurrence with polynomial coefficients.

Table 2. Weight-5 families $\sum \binom{n}{k}^a \binom{n+k}{k}^b$.

a	b	Recurrence order	Growth rate	Status
2	2	2	≈ 27.23	Apéry numbers; known
3	1	≥ 8	≈ 17.47	Not found
4	**1**	**4**	$\approx$ **43.04**	S_{20}; **this paper**
5	0	≥ 8	≈ 59.68	Not found
3	2	≥ 8	≈ 47.59	Not found
2	3	≥ 8	≈ 83.02	Not found
1	4	≥ 8	≈ 201.57	Not found

Remark 5.1. Among all weight-5 partitions, S_{20} is the *only new sequence* (beyond the known Apéry-type $(2, 2)$ case) for which a finite recurrence has been established. This suggests that the $(4, 1)$ partition occupies a special position in the hierarchy. The remaining partitions may satisfy recurrences of order ≥ 8, but our search with 30 exact terms and polynomial degree bounds up to 9 did not produce them.

6. Mirror Map Integrality

Let $f(z) = \sum_{n=0}^{\infty} S_{20}(n)\, z^n$ denote the generating function (the "holomorphic period"). Following the Calabi–Yau paradigm [6, 2], define the logarithmic period

$$g(z) = \sum_{n=0}^{\infty} S_{20}(n)\, H(n)\, z^n$$

where $H(n)$ encodes an appropriate harmonic-number combination, and the *mirror map*

$$q(z) = z\, \exp\bigl(g(z)/f(z)\bigr) = z + \sum_{d=2}^{\infty} q_d\, z^d. \tag{7}$$

Theorem 6.1 (Mirror map integrality). *All coefficients q_d of the mirror map* (7) *associated with $S_{20}(n)$ satisfy $q_d \in \mathbb{Z}$ for $d \leq 16$.*

Proof (computational). The coefficients q_d for $d \leq 16$ were computed using exact rational arithmetic (no floating-point approximation) and verified to be integers. □

The computed values are:

Conjecture 6.2 (Lian–Yau integrality). *The mirror map coefficients $q_d \in \mathbb{Z}$ for* all *$d \geq 1$.*

If true, this would place S_{20} alongside the Apéry numbers and the 14 Calabi–Yau operators catalogued by Almkvist–van Enckevort–van Straten–Zudilin as sequences with the Lian–Yau integrality property.

7. The Diagonal Representation

A key question in arithmetic combinatorics is whether a D-finite integer sequence can be represented as the diagonal of a multivariate rational function in a minimal number of variables. For the Callens-Schmidt sequence $S_{20}(n)$, we prove that it is the diagonal of an asymmetric 5-variable rational function:

Theorem 7.1 (Diagonal representation). *Let $I(x_1, x_2, x_3, x_4, x_5)$ be the rational function*

$$I(x_1, x_2, x_3, x_4, x_5) = \frac{1}{(1 - x_1)(1 - x_2)(1 - x_3)(1 - x_4)(1 - x_5) - x_1 x_2 x_3 x_4}.$$

TABLE 3. Mirror map coefficients q_d for $d \leq 16$.

d	q_d
2	9
3	165
4	4 110
5	111 075
6	3 316 785
7	104 271 733
8	3 421 974 692
9	115 918 914 756
10	4 027 088 171 898
11	142 793 489 195 634
12	5 149 415 166 799 466
13	188 353 171 046 524 999
14	6 973 330 284 143 733 181
15	260 877 511 906 858 891 334
16	9 848 682 801 654 949 278 015

Then the Callens-Schmidt sequence $S_{20}(n)$ is its diagonal:

$$S_{20}(n) = [x_1^n x_2^n x_3^n x_4^n x_5^n]\, I(x_1, x_2, x_3, x_4, x_5).$$

Proof. Let $D(x_1, \dots, x_5) = (1 - x_1)(1 - x_2)(1 - x_3)(1 - x_4)(1 - x_5)$ and $M(x_1, \dots, x_5) = x_1 x_2 x_3 x_4$. We expand the rational function I as a geometric series in M/D:

$$I(x_1, \dots, x_5) = \frac{1}{D - M} = \sum_{r \geq 0} \frac{M^r}{D^{r+1}} = \sum_{r \geq 0} \frac{(x_1 x_2 x_3 x_4)^r}{(1 - x_1)^{r+1}(1 - x_2)^{r+1}(1 - x_3)^{r+1}(1 - x_4)^{r+1}(1 - x_5)^{r+1}}$$

To extract the diagonal coefficient $[x_1^n \cdots x_5^n]$, we expand each $(1 - x)^{-(r+1)}$ using the Taylor series

$$\frac{1}{(1 - x)^{r+1}} = \sum_{m \geq 0} \binom{m + r}{r} x^m.$$

For each variable:

- For $i \in \{1, 2, 3, 4\}$, the numerator contributes x_i^r, so we must extract the coefficient of x_i^{n-r} from $(1 - x_i)^{-(r+1)}$. This is given by $\binom{(n-r)+r}{r} = \binom{n}{r}$.
- For x_5, there is no numerator contribution, so we must extract the coefficient of x_5^n from $(1 - x_5)^{-(r+1)}$. This is given by $\binom{n+r}{r}$.

Multiplying the contributions for each variable and summing over r gives:

$$[x_1^n \cdots x_5^n]\, I(x_1, \ldots, x_5) = \sum_{r=0}^{n} \binom{n}{r}^4 \binom{n+r}{r}.$$

By renaming the summation index r to k, this is exactly the definition of $S_{20}(n)$ given in Equation (2). $\square$

Remark 7.2 (Symmetric vs. Asymmetric form). The naive symmetric candidate

$$\frac{1}{(1 - x_1 - x_2 - x_3 - x_4)(1 - x_5) - x_1 x_2 x_3 x_4 x_5}$$

does *not* yield $S_{20}(n)$; its diagonal evaluates to exactly 2^n (Callens, 2026). The asymmetric form of Theorem 7.1 breaks the symmetry but successfully represents the sequence using the minimal number of variables (5).

Remark 7.3 (Calabi-Yau 3-Fold Geometry). In mirror symmetry, the dimension of the Calabi-Yau manifold is linked to the order of the minimal Picard-Fuchs equation. Because the minimal Picard-Fuchs operator has order 4, the underlying geometry of $S_{20}(n)$ is that of a Calabi-Yau 3-fold motive, rather than a 4-fold as the weight-5 binomial structure might initially suggest. This drop in weight is a key geometric feature of the sequence.

8. Lean 4 Verification

As a partial formal check, the identity (6) (the recurrence at $n = 0$) has been compiled in the Lean 4 proof assistant:

```
def S20 (n : Nat) : Int :=
  ((Finset.range (n + 1)).sum
    (fun k => (Nat.choose n k)^4
            * (Nat.choose (n + k) k)))

theorem theorem20_inst0 :
    (-5412650858431135013634958175726842170573378411840)
      * S20 0
  + (-6600211789894833600749251782579095561783149274990400)
      * S20 1
  + (-2972423453762967355073866981445913843111540130320624O)
      * S20 2
  + (-667529688600156302761716408138316739499698559647824O)
      * S20 3
```

```
+ (-272198721521932617277293245047721130052020296806560)
    * S20 4
+ (20478134952232355172884134183653971676016433020000)
    * S20 5
= 0 := by decide
```

Remark 8.1 (Scope of the Lean verification). The Lean 4 theorem `theorem20_inst0` verifies a *single numerical identity*: the linear combination of $S_{20}(0), \ldots, S_{20}(5)$ with the given integer coefficients equals zero. This confirms the recurrence at the point $n = 0$ but does *not* constitute a proof of the recurrence for all n. A full formal proof would require either formalising the creative telescoping certificate or establishing a uniform inductive argument, which remains an open problem.

9. Open Questions

We collect the principal open problems arising from this work.

(1) **Lian–Yau integrality.** Is $q_d \in \mathbb{Z}$ for all $d \geq 1$? (Conjecture 6.2.) A proof would likely require a geometric construction (e.g., a mirror pair of Calabi–Yau threefolds).
(2) **Full formal verification.** Can the minimal order-4 recurrence (or its order-5 left-multiple) be proved for all n in a proof assistant, either by formalising the creative telescoping certificate or by an independent method?
(3) **Supercongruences.** Do there exist p-adic supercongruences for S_{20} analogous to those known for Apéry numbers? The $\frac{3}{4}$-well-poised structure may impose constraints on which primes exhibit enhanced congruences.
(4) **Modularity.** Is the generating function $f(z) = \sum S_{20}(n)z^n$ a modular form or a period of a modular motive? The order-5 recurrence operator should be analysed for G-operator properties in the sense of André.
(5) **Remaining weight-5 partitions.** What are the recurrence orders for the partitions $(3,1)$, $(5,0)$, $(3,2)$, $(2,3)$, $(1,4)$? Are they truly of order ≥ 8, or do they require higher-degree polynomial coefficients?
(6) **Picard–Fuchs operator.** The Picard–Fuchs ODE has singularities at $t \approx 0.0232, -0.0692, -0.0736 \pm 0.0355i, -238.8$. The last singularity is far from the origin. Is it an apparent singularity, and does the operator factorise over $\mathbb{Q}(t)$?

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